

Curriculum Vitae

Divakara G R

+91-9901590775

divakar.gowda.r@gmail.com

Objective

7+ Years Of experience in Embedded C coding and debugging. Worked On CAN, UART, SPI and I2C protocols and WiFi, Worked On STM32, 8051 micro-controllers using keil u vision IDE and ESP32 using esp-idf.

Professional Summary

- Experience in Automotive ADAS features implementation in RADARS and automotive tools and testing ADAS features and debugging the bugs in ECUs.
- Experience in Embedded C coding, Debugging and work experience on Micro-controller(STM32 and ESP32 using esp-idf).
- Experience in implementing Embedded C code using FREERTOS APIs and Inter task communication mechanisms(Using Queues and mutex).
- Experience in implementing power management in programming by making all the peripheral in sleep mode and calculating power consumption.
- Delivered all the requirements On time(Completed all requirements within the timeline).
- Experience in Micro-controller and peripheral interfacing using different protocols(SPI, I2C, UART).
- Experience in QUAD SPI and CAN Communication
- Experience in UDS protocols.
- Experience in Client Communication and Status Reporting.
- GOOD written and Oral communication skills.

Professional Experience

- Working as Senior software engineer in Aptiv automotive from Aug 2021.
- Worked as a Firmware Engineer in L&T Technology Services Limited from DEC 2018 to Aug 2021
- Worked as Junior firmware Engineer in Avanzati Product Technologies, Mysore from Apr 2017 to Mar 2018.
- Worked as Junior firmware Engineer in Pecsol Technologies, Mysore from Jan 2016 to Mar 2017.

Skills

Domain	Automotive, Embedded.
Programming Skills	C, Embedded C, Cpp, FreeRTOS APIs, Linux concepts and socket programming.
Client Tools	DOORs and JIRA.
Software	TRACE32, CANoe, Wireshark. ESP-idf, STMCube, Eclipse, Keil
Technical skills	Experience on Communication protocols like CAN, UART, I2C, SPI StandardSPI and Quad SPI.

Projects (continued)

2012 to 2015 ■ ■ BE in Electronics and Communication, 2015, Mysore.

2009 to 2012 ■ Diploma in Electronics and Communication, 2012, Mysore.

Projects

Project-1 ■ Short range Radar(SRR).

■ Design

- **Duration:** Sep 2021 to On going
- **Team:** 4
- **Tools:** Trace32, Canoe, CAN protocol.
- **Description:** Implementation of UDP and Features for the ADAS system to assist driver and Problem resolution for the bugs reported in vehicle test.

■ Role: Senior engineer

■ Responsibilities:

- To get the code implementation from the other team and integrating with the existing code by making the necessary changes required in Wrapper.
- Building the entire program and validating all the features
- Adding the feature signals to UDP and CAN logging
- Validating all the features in both CAN and UDP.
- If any problems reported analyzing the resolution for the problem and fixing it.

Project-2 ■ Heater(Air and water heater) control from HMI.

■ Design

- **Duration:** Sep 2020 to On going
- **Team:** 3
- **Tools:** STM32CubeIDE, esp-idf, CAN, SPI, UART, FREERTOS.
- **Description:** Controlling Heater Operations, ON / OFF Or setting Temperature and mode from HMI

■ Role: Firmware Engineer

■ Responsibilities:

- To send commands from HMI to heater over CAN messages, by creating the CAN messages.
- Sending the CAN messages by creating CAN rx/tx task using FREERTOS in ESP8266 Controller.
- Controlling the Heater Operations from HMI using the external buttons interfaced to STM32 Controller with respect to external interrupt.
- Controlling the HMI Display using STM32 Controller with respect to files generated from script written to create array for images and navigation.
- Writing the input files(.txt) for python script to generate, files required by STM32 Controller (array for images and navigation).

Projects (continued)

- Project-3** **Flashing STM32 from hex file using inbuilt boot-loader commands.**
- **Design**
- **Duration:** Apr 2020 to May 2020
 - **Team:** 3
 - **Tools:** STM32CubeIDE, real-term.
 - **Description:** Writing the hex file contents into internal flash of STM controller using inbuilt boot-loader commands.
- **Role:** Firmware Engineer
- **Responsibilities:**
- To transmit generated hex file for simple led blinking code for STM.
 - Writing the hex file contents into internal flash of STM using boot-loader commands.
 - Starts executing from the recent update.
- Project-4** **Working on RVC library implementation on ESP32**
- **Design**
- **Technologies:** CAN and Embedded-C and AWS cloud concepts
 - **Duration:** Mar 2020 to Apr 2020
 - **Team:** 3
 - **Tools:** ESP-idf tool chain.
 - **Description:** Implementation of RV-C CAN messages in ESP32.
- **Responsibilities:**
- To create CAN messages with respect to Data group number(DGN) as CAN identifier(extended id) using FREERTOS APIs in ESPSR.
 - To transmit data or receive data from devices to ESP via CAN.
 - Push the data to cloud using MQTT protocol.
 - Inter process communication for get data and transmit.
- Project-5** **Cloud Communication and Power management implementation on ESP32**
- **Design**
- **Technologies:** Embedded C, WiFi
 - **Duration:** OCT 2019 to Jan 2020
 - **Team:** 3
 - **Tools:** ESP-idf.
 - **Description:** Collecting data from different Devices and sending data to the cloud and Implementing power management when no devices were active On ESP32.
- **Role:** Firmware Engineer
- **Responsibilities:**
- Implementing code to push data to cloud using FreeRTOS APIs and AWS Shadow APIs via WiFi.
 - Analyzing datasheets of ESP32 to explore different power saving modes.
 - Implementing different power saving modes in ESP32 to make the whole system as power efficient.
 - Controlling the ESP32 with respect to data from devices and push data to cloud to make the system as power efficient.
 - Controlling all the peripherals connected to ESP32 by activating their sleep and active modes from ESP32.

Projects (continued)

Project-6 Writing External Flash from STMSR

Design

- **Technologies:** Quad SPI protocol and Embedded C
- **Duration:** Jan 2020 to Feb 2020
- **Team:** 3
- **Tools:** STM32CubeIDE.
- **Description:** Writing Binary file contents into external flash WRUQ80DV from STM32 using Quad-SPI interface and YMODEM.

Role: Firmware Engineer

Responsibilities:

- To develop Flash API drivers.
- Need to store binary data/graphics data from the PC to Winbond flash via STM32.
- Perform the timing analysis with respect to CPU cycles needed to store and retrieve the data from Winbond Flash.

Project-7 Quad copter (previous company project) Using STM32 micro-controller

Design

- **Technologies:** Embedded C and SPI protocol
- **Team:** 10
- **Tools:** STMcube, Keil u-vision
- **Description:** Quad copters are mainly used for skydiving, it consists of camera and GPS.

Role: Firmware Engineer

Responsibilities:

- Developing code to interface micro-controller and IMU sensor used for quad copter.
- Analyzing datasheets of both IMU sensor and micro-controller.
- Interfacing IMU with micro-controller using SPI protocol.
- Analyzing data received from IMU and converting received data (Hex value) to normal float value.
- Separating received data as Roll, Yaw and Pitch values.

Place:
Bengaluru.

Divakara G R

Date:

Signature